

Exploring the nexus between transit-based job accessibility and labor market outcomes among marital immigrants

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ABSTRACT

Foreign spouses often face disadvantages which may directly impede their ability to gain and retain employment. This research investigates the role of transit-based job accessibility in overcoming employment barriers among marital immigrants in Taiwan. Leveraging government survey data, we develop multilevel econometric models to disentangle the differential impact of job accessibility on labor market outcomes across metropolitan (Taichung City) and regional (Central Taiwan) settings. We find that better transit-based job accessibility positively influences the employment status and likelihood of above-minimum-wage earnings among foreign spouses in Taichung City. In Central Taiwan, better accessibility is associated with higher income and an increased likelihood of above-minimum-wage earnings. These results challenge the notion that improving accessibility offers a silver bullet to tackle all issues related to social equity. Instead, we highlight the necessity of targeted interventions to address the multifaceted needs of marital immigrants, with implications aimed at fostering economic integration and social inclusion.

1. Introduction

The spatial divide between residential and employment locations can present substantial barriers to economic participation, particularly for marginalized groups who may lack the means or motivation to overcome these challenges. For instance, individuals with limited mobility may confine their job searches to nearby opportunities that may be fewer, lower-paying, or misaligned with their qualifications (Phillips, 2014). This constraint is especially evident in areas where employment is concentrated in specific sectors, such as low-wage, primary-sector jobs. In such cases, space not only limits the range of job choices but may also diminish potential for upward mobility.

These spatial barriers manifest most acutely when transport systems fail to connect people to opportunities. In transport geography, the concept of accessibility—the ease of reaching desired destinations, including workplaces—provides a perspective for understanding how transport and land use systems can potentially exacerbate or help overcome these challenges (Guzman et al., 2017; Hansen, 1959; Pereira et al., 2017). Poor accessibility increases the risk of transport-related social exclusion, a condition wherein inadequate transport services further marginalize disadvantaged populations by isolating them from

jobs, services, and social participation (Kenyon et al., 2002; Lucas, 2012). In contrast, improved accessibility has the potential to expand job search areas, increase employment prospects, and improve labor market outcomes. A substantial body of research has empirically explored how accessibility to jobs, alongside individual socio-economic factors, may, to some extent, explain employment status, salary levels, and job retention (Bastiaanssen et al., 2020).

Taiwan and its growing population of marital immigrants (i.e., foreign-born individuals who migrate through transnational marriage) present an important yet underexamined case for examining transport accessibility and labor market dynamics in a non-Western setting. As de facto outsiders, marital immigrants occupy a marginal societal status—neither fully integrated citizens nor temporary migrants—which may create distinctive labor market vulnerabilities that remain underexplored in transport geography literature.

The majority of marriage immigrants in Taiwan are foreign brides from Mainland China and Southeast Asia who marry men of lower socio-economic statuses, often with expectations that they will prioritize household responsibilities (Tsai, 2011). Although many of these individuals express a strong commitment to joining the workforce, their employment prospects are often constrained by a limited time budget,

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lower educational attainment, and language barriers (Kim et al., 2020; National Immigration Agency, 2020). These challenges exist in addition to pervasive social biases that frame them as outcasts and, in more extreme cases, as an “inferior” social class (Chen and Liu, 2017; Hsia, 2007; Tsai, 2011). As a result, employers in Taiwan tend to view marital immigrants as a supplementary workforce—an option only when local candidates are unavailable or unwilling to take on certain jobs (Kim et al., 2020). Even when they do find employment, marital immigrants are often relegated to lower-level positions in specific sectors (Kim et al., 2020). Lower incomes make private vehicle ownership less feasible and those who live in areas with limited transit options may find that opportunities are largely out of reach.

For marital immigrants in Taiwan, having decent levels of baseline accessibility to nearby employment can help them secure jobs that fit within their time constraints and reduce the need to conduct job searches in unfamiliar territories where they may lack local knowledge and networks (Gobillon et al., 2011; Gobillon et al., 2007). However, proximity to jobs alone may not always suffice. In some cases, nearby positions may not align with their educational background or language skills, or nearby employers may exhibit prejudice against hiring immigrants. In addition, competition may play a substantial role—being close to job opportunities may be of little advantage if other, more qualified jobseekers also live nearby and compete for the same positions (Parks, 2004). In these contexts, greater accessibility through transit connectivity can encourage more extensive job searches (Phillips, 2014), thereby broadening the range of available jobs or potentially open up opportunities in less congested parts of the city where competition is lower. Such mechanisms are not confined to those aiming to enter the workforce. Employed foreign spouses can leverage better choices to switch to higher-paying positions, or help them transition out of informal, part-time work and secure jobs that offer sufficient hours to earn above the minimum wage. We hypothesize that transit-based job accessibility can help offset some of the structural disadvantages faced by marital immigrants and provide them with a stronger motivation and means to overcome spatial barriers and engage more effectively in the labor market.

Despite a growing body of research on accessibility and individual employment outcomes, several empirical and methodological gaps remain. First, most studies have been conducted in Western urbanized contexts (Hernandez et al., 2020). The basis for much of this research is the *spatial mismatch hypothesis* (SMH) (Kain, 1968), which was developed in the socio-economic setting of the United States to address issues such as racial segregation and suburbanization. However, the socio-spatial dynamics that affect Taiwan’s labor market, transport and land use systems, and immigrant population substantially differ. Taiwan’s urban areas, like those of other developed East Asian cities, are generally denser and more mixed-use, with comparatively lower parking provision (Barter, 2011; Chen et al., 2020). These conditions encourage relatively higher public transport use instead of car dependency. Taiwan’s economy also retains a strong manufacturing base relative to many deindustrialized Western regions which have shifted towards finance, technology, or other service-based sectors (Autor et al., 2016; Hsiao, 2013; Rodrik, 2016). This industrial orientation means that jobs are not exclusively concentrated in service-sector downtowns, but may also be located in dispersed factories and industrial districts. Further, Taiwan’s labor market has seen an increase in non-standard job arrangements (e.g., part-time and informal employment) in recent years, many of which are held by foreign-born workers (Hsiao, 2013). Thus, the findings and mechanisms from U.S.-based studies may not be directly applicable to Taiwan’s context.

In addition, most studies focus on minority populations in large metropolitan areas, leaving unanswered questions regarding potential heterogeneous effects of job accessibility when extending analyses to regional or less urbanized settings. In Taiwan, metropolitan and outlying areas differ substantially in land use and transport infrastructure. Taiwanese cities often have higher job density and comprehensive

public transit systems that jobseekers can use to reach a wider range of employment opportunities across various locations. In contrast, regions outside major urban areas typically feature more dispersed job sites, fewer transit options, and a concentration of primary-sector employment. These contextual differences suggest that the opportunities and constraints jobseekers face may vary; therefore, the notion that transit-based job accessibility may not offer a one-size-fits-all solution to improving outcomes would require more investigation.

From a methodological perspective, most existing studies use single-level econometric models. These models typically combine accessibility estimates (i.e., area-level variable) with individual-level factors, which may fail to account for the hierarchical structure of survey data. These approaches may thus lead to underestimated standard errors and consequently, an increased risk of false positives in the relationships explored (Dupont et al., 2013; Park et al., 2017).

This paper seeks to address these gaps by investigating the impact of job accessibility on labor market outcomes among foreign spouses in Taiwan. We focus on transit-based job accessibility, or the ability to reach job destinations through public transit. Our explicit focus on transit is motivated by equity implications, particularly for marginalized individuals who may not have the means of owning automobiles. We frame our research around the following questions:

- Is better transit-based job accessibility associated with improved labor market outcomes for marital immigrants in Taiwan?
- How does the impact of transit-based job accessibility differ between metropolitan and regional settings?

We first calculate competitive transit-based job accessibility levels at the smallest administrative level for which data is available. We then combine the accessibility measures with individual-level data from a survey conducted among foreign spouses. We developed multilevel econometric models to estimate the impact of transit-based job accessibility on three individual labor market outcomes: employment status (H1), income (H2), and likelihood of earning above the minimum wage (H3). For each outcome, we estimated separate models for metropolitan and regional settings, resulting in six main models. We contribute to the literature by examining an underexplored socio-spatial context, providing evidence of the differential effects of accessibility across metropolitan and regional settings, and proposing methodological improvements.

The remainder of the paper is structured as follows: Section 2 provides a deeper contextual understanding of marital immigrants in Taiwan and their disadvantage, as well as a review of relevant literature on transport and individual employment outcomes. Section 3 outlines the theoretical foundations for our hypotheses. Section 4 describes the study area, data sources, and variables used in our analyses. In Section 5, we discuss our methodological approach. Section 6 presents and discusses the estimation results. Finally, Section 7 offers conclusions, policy implications, and directions for future research.

2. Literature review

2.1. Marital immigrants and disadvantage

The circumstances of foreign spouses in Taiwan represent a distinct immigration niche compared to immigrants in industrialized nations. In countries such as the United States, immigrants often arrive with close family members or settle within ethnic communities which could provide social networks to support job searches (Portes et al., 2002). These social networks can help them enter specific occupations (Borjas, 2006) and may, in some cases, compensate for poor spatial access to jobs (Parks, 2004). In contrast, foreign spouses in Taiwan typically arrive alone and are widely dispersed across towns and cities (Tsai, 2011), as their place of residence is often predetermined by their spouse’s location. This geographical dispersal may limit their ability to form strong

social networks, leaving them potentially disadvantaged in gaining information about job openings through informal channels such as word-of-mouth (Calvó-Armengol and Zenou, 2005).

Foreign spouses face additional challenges due to the disadvantaged statuses of the families that they marry into. Transnational marriages emerged largely as a response to a shortage of single women, particularly in rural areas. Since the late 1970s, Taiwanese men of lower socio-economic status sought brides from Southeast Asia through marriage brokers (Chen, 2008; Hsia, 2000; Wang and Chang, 2002). Inter-marriage between Taiwanese and Mainland Chinese individuals also increased after the martial law was lifted in 1987 (Tsai, 2011; Tsai, 2023). Taiwan's economic growth and increased educational levels among Taiwanese women further reduced marriage prospects for men of lower socio-economic standing, as many Taiwanese women began seeking partners with similar or higher education and economic capital (Chen and Liu, 2017; Chen, 2008). This is consistent with the patriarchal norms of East Asian society, where men are expected to “marry down” and women to “marry up” (Hsia, 2000; Kim et al., 2020). As a result, men working in lower-income sectors increasingly turned to marriage brokers to “purchase” wives from abroad (Chen and Liu, 2017). Within these families, marital immigrants are often regarded as ‘tied movers,’ who are expected to prioritize household responsibilities and act as secondary contributors to household income (Chuang et al., 2010).

At an individual level, immigrants face limited prospects to work partly due to lower or undervalued human capital. Most foreign spouses received mid-level education (i.e., middle or high school/vocational training) or less before arriving in Taiwan, with education levels typically lower among Southeast Asian immigrants (National Immigration Agency, 2020; Tsai, 2011). A study found that education had a positive effect on the employment status of native-born women in Pingtung County, Taiwan, but was largely irrelevant for Southeast Asian foreign brides (Chen and Liu, 2017). This suggests that the labor market largely ignores the human capital of marital immigrants (Chen and Liu, 2017). The skills acquired in their home countries may not be recognized in Taiwan, and as such, many of them abandon their prior competencies and shift to new professions (National Immigration Agency, 2020).

Another important factor is language proficiency. Proficiency in Chinese is important for the day-to-day operations in Taiwanese companies, especially those involving customer interactions (Chuang et al., 2010; Kim et al., 2020). Many employers in Taiwan thus expect immigrants to take language lessons (Kim et al., 2020). While Southeast Asian immigrants, especially those without Chinese heritage, face the greatest disadvantages (Tsai, 2011), even Mainland Chinese spouses may encounter biases against their slightly different accents, which may cause employers to hesitate hiring them despite their ability to communicate with locals fluently (Chuang et al., 2010).

These structural barriers exist in addition to anti-immigrant attitudes. Foreign brides have often been portrayed by the media as materialistic or prone to criminal conduct, while the Taiwanese men who marry them have been labelled as socially undesirable (Hsia, 2007). These stereotypes serve to perpetuate the notion that marriage immigrants are not only of lower social status, but are also racialized as an inferior group and a potential “threat to the nation” (Hsia, 2007; Tsai, 2011). Southeast Asian marriage migrants, in particular, have been described as “fertile brides” but “culturally unfit mothers” for raising the next generation, given their lack of familiarity with the local language and culture (Lan, 2008). These biases extend to their children, with public concerns that an increasing number of children born to foreign brides could lower the “average population quality” of Taiwan (Hsia, 2007; Tsai, 2011). In 2006, one legislator went so far as to publicly claim that many children born to Vietnamese mothers might suffer from birth defects due to chemical exposure during the Vietnam War and warned the government against allocating resources to immigrant families who could face health issues (Hsia, 2007).

This social construct of marriage immigrants as “outsiders” carry over to the labor market. Employers tend to view foreign spouses as a

supplemental labor force, hiring them only when local workers are unavailable (Kim et al., 2020; Wang, 2001). In some cases, even when marriage immigrants possess the required skills and language proficiency, employers may remain reluctant to hire them unless they are referred by a family friend or relative (Kim et al., 2020). These dynamics are further complicated by ethnic politics, especially for Mainland Chinese spouses, who may face exclusionist attitudes in line with Taiwan's evolving national identity vis-à-vis China (Tsai, 2011). According to the National Immigration Agency (2020), Mainland Chinese spouses report higher rates of difficulty in securing employment, largely due to workplace discrimination and strict legal restrictions that historically disallowed them from entering the labor market for their first two years of residence. Even after this period, they are only granted work permits under highly restrictive conditions (Chuang et al., 2010; Tsai, 2011).

In sum, marital immigrants in Taiwan face a unique set of challenges related to weak social networks, undervalued education and skills, language barriers, and societal prejudice. Despite these obstacles, foreign spouses are often highly motivated to work. As of 2018, their labor force participation rate exceeded that of the Taiwanese population (National Immigration Agency, 2020). However, they remain confined mostly to lower-grade jobs and sectors, with marital immigrant families earning roughly half the national average income of Taiwanese families (Kim et al., 2020; National Immigration Agency, 2020; Tsai, 2011). This is despite their longer working hours and higher rates of job-related injuries (Kim et al., 2020).

These socio-economic disadvantages are closely linked to transport disadvantage. Lower incomes mean many immigrant families cannot afford private vehicles, and those who rely on inadequate public transit may thus be isolated from employment opportunities. For example, Parks (2004) demonstrated how spatial accessibility could improve employment outcomes for specific immigrant groups in the U.S. However, unlike immigrants in the U.S. who typically migrate for labor opportunities and must often demonstrate their capacity to contribute to the workforce, foreign spouses in Taiwan primarily migrate for family reasons. This raises the strong possibility that foreign spouses may face an even more severe level of exclusion, as they are not initially positioned as economic contributors. This study seeks to address this gap by providing new evidence on the impact of transit-based job accessibility for this underrepresented demographic in a non-Western context.

2.2. Transport accessibility and individual labor market outcomes

A substantial body of research has examined the relationship between transport and employment outcomes, often within the framework of the spatial mismatch hypothesis. There remains no singular consensus among empirical evidence, in large part due to the variations in methodologies and the distinct socio-spatial contexts in which these studies have been conducted (Bastiaanssen et al., 2020; Hu, 2017; Sandoval et al., 2011).

Earlier studies primarily relied on proximity-based metrics, such as distance to transit stops (Sanchez, 1999) or the number of transit stations within a certain radius (Ong and Houston, 2002). However, proximity alone may not capture the overall impact of transport on individuals' ability to reach opportunities, as it fails to consider the system at the network level. In other words, living near a transit stop is not advantageous if it does not guarantee an individual to reach desired destinations within a reasonable time frame. Moreover, the impact of distance from a transit stop can vary greatly depending on the context. A person living 100 m away from a transit stop in a suburban area may not have the same capacity to reach as many job opportunities as a person living 100 m away from a transit stop in a central business district.

Recent literature has shifted towards using job accessibility as a more comprehensive metric to better understand the nexus between transport and labor market outcomes. Job accessibility is broadly defined as the potential extent to which individuals can effectively reach employment opportunities (Guzman et al., 2017; Hansen, 1959). Accessibility

captures the level of integration of land use and transport systems. Proximity to opportunities and connectivity via transport each contribute to accessibility and these elements can also partially compensate for one another. For instance, strong connectivity may offset limited nearby job options, while a high density of nearby jobs can mitigate limited transport options. In a broader sense, accessibility reflects more than just physical reach; it also serves as a proxy for freedom of choice—the range of destinations that an individual can realistically pursue, and by extension, the aspirations that they can achieve (Martens, 2012).

In this section, we review studies which used job accessibility as a metric to examine the impact of transport on individual labor market outcomes. We exclude studies that rely on other proxies for transport, such as proximity-based measures or automobile ownership. Similarly, research analyzing job accessibility and employment outcomes at broader spatial levels (i.e., census tracts) falls beyond the scope of this section.

Given the origins of the SMH in the United States, much of the empirical evidence have been derived from American cities. U.S. urban development patterns typically exhibit higher car dependency, job suburbanization, and residential segregation by income and race (Hernandez et al., 2020; Kain, 1968). Findings from the U.S. demonstrate mixed results. For example, some studies show that higher ratios of public transport to car job access (Kawabata, 2003) and better bus job access (Yi, 2006) are associated with better employment prospects. In contrast, Sandoval et al. (2011) found that spatial accessibility to jobs was less relevant for employment outcomes among welfare recipients in three Californian counties, whereas car ownership emerged as a more significant factor. Bania et al. (2008) found no significant effects of car- or bus-based job accessibility on employment among welfare recipients in Ohio. Blumenberg and Pierce (2014) and Blumenberg and Pierce (2017) demonstrated using longitudinal data that automobile ownership appeared to be a more significant determinant of employment status and job retention among welfare recipients across metropolitan areas than public transit accessibility. Hu (2017) found a significant positive association between car-based job accessibility and employment among medium to low-income groups in Los Angeles, though this effect did not hold across other income brackets. Blumenberg et al. (2019) further showed that access to public transit is an important factor in securing employment, particularly for low-income older adults in carless households.

While these studies contribute to our understanding of accessibility's role in explaining employment outcomes, the distinct land use patterns, transport systems, and socio-demographic characteristics across regions suggest that U.S.-based conclusions may not be universally applicable. For example, European and Asian cities typically feature denser urban forms and higher reliance on public transit, walking, or cycling for commuting to work. Policy recommendations from U.S. contexts, such as promoting car ownership among low-income groups, may have less relevance in these settings. As reflected by the lack of consensus from U.S.-based studies, job accessibility alone is unlikely to be a silver bullet for addressing labor market disparities and context-sensitive approaches would be needed.

Studies in Europe generally identified job accessibility as a crucial factor in the employment prospects among disadvantaged groups. For instance, poor job accessibility has been linked to reduced employment prospects among low-skilled jobseekers in France (Korsu and Wengleski, 2010) and females in Spain (Matas et al., 2010). Di Paolo et al. (2017) found that public transport accessibility significantly influenced female employment in Spain, with an insignificant effect on male employment. In Great Britain, Bastiaanssen et al. (2022) found that public transport job accessibility positively affects employment outcomes, but only in areas with lower car ownership rates and lower-income neighborhoods. In the Netherlands, Bastiaanssen et al. (2025) also found significant positive effects of combined public transport and bicycle job accessibility on employment.

In recent years, studies have also emerged in Latin American cities, which often exhibit a core-periphery urban structure with concentrated employment opportunities in city centers and vulnerable populations residing in the urban periphery (Hernandez et al., 2020). Long commutes and the need to rely on public transit are therefore among the barriers to employment among the urban poor (Hernandez et al., 2020; Oviedo Hernandez and Titheridge, 2016). In Uruguay, Hernandez et al. (2020) found that public transport job accessibility is associated with a lower probability of unemployment. According to the findings of Quintanar (2015) in Mexico City and Duarte et al. (2023b) in Sao Paulo, job accessibility has a more significant impact on the employment prospects of women compared to men.

In Taiwan, Lin et al. (2016) examined the effect of job accessibility on employment for urban and rural aborigines and found no evidence of a significant relationship. However, the study's small sample sizes may have constrained the model's power and limited the ability to fully capture the relationship. In Ghana, Chen et al. (2017) found a significant positive association between job accessibility and the likelihood of employment.

Compared to the extensive literature linking job accessibility to employment probability, research on job accessibility and salary outcomes remains limited. Bania et al. (2008) found no significant association between car-based job accessibility and earnings in Ohio, while Lin et al. (2016) found a positive correlation between job accessibility and salaries for Taiwanese aborigines. Studies from Brazil also found positive relationships between better job accessibility and higher wages (Haddad and Barufi, 2017), as well as reduced likelihood of low-wage employment (Duarte et al., 2023a). To the best of the authors' knowledge, no existing studies have specifically explored the relationship between job accessibility and an individual's likelihood of earning above the minimum wage.

Disparities in empirical results can also be attributed to the differences in how studies measure job accessibility. Many studies overlook competition in their accessibility metrics. Accessibility measures may yield biased results if they fail to account for the finite nature of job opportunities relative to demand (Higgins et al., 2022; Parks, 2004). Research suggests that incorporating demand-side adjustments leads to more accurate predictions of employment outcomes (Merlin and Hu, 2017).

In addition, many studies use single-level econometric models, which integrate area-level accessibility metrics with individual-level data. These do not account for the hierarchical structure of survey data and may introduce bias by potentially violating the assumption of independent residuals in generalized linear models. Survey respondents are often clustered within spatial units and we can assume that individuals within the same spatial unit may be more similar in terms of characteristics and outcomes than those in other spatial units. Single-level econometric models may thus risk underestimating standard errors and produce overly narrow confidence intervals, which can lead to an inflated risk of Type I errors (Dupont et al., 2013; Park et al., 2017). In other words, these models may suggest that accessibility has significant effects on employment when such effects may not exist. To our knowledge, Hernandez et al. (2020) is the only study to apply multilevel models, though their study did not account for competition among jobseekers in their accessibility metrics.

3. Hypotheses development

We propose four hypotheses to be tested using individual-level data from a nationwide survey conducted among foreign spouses in Taiwan.

H1. *Transit-based job accessibility positively affects employment status.*

In terms of employment status, we hypothesize that an increase in job opportunities that are easily reachable from a marital immigrant's residence will have a positive impact on their employment status. We explain several mechanisms which may underpin this hypothesis,

considering that transit-based job accessibility, in the context of this paper, is largely determined by proximity to job opportunities, connectivity via transit, and competition for available jobs.

First, marital immigrants may benefit from having decent levels of baseline accessibility to nearby jobs. Proximity to work means less time for commuting and more time that can be dedicated to engaging in the labor market. This aspect is important for foreign spouses, who often shoulder greater responsibilities at home, such as childcare or caring for in-laws (Chuang et al., 2010). The concept of *social entrapment* (Hanson and Pratt, 1994) is particularly relevant, as foreign spouses, especially women, may have limited time to commute due to domestic duties, which may constrain their job-search area and consequently restrict the range of employment opportunities available to them (Parks, 2004). Nearby job opportunities make it easier for them to manage work and family duties better, thereby strengthening their motivation to potentially cross the threshold from unemployed to part-time employed status. As childcare responsibilities decrease, such as when children grow older and become more independent, foreign spouses may also fully re-enter the labor market (i.e., transition from part-time to full-time status) amidst increasing financial pressure to contribute to household income (Lu, 1992). From employers' perspective, companies may favor candidates with shorter commutes, as this can enhance punctuality and reduce work-family conflicts that may affect productivity (Gobillon et al., 2011; Gobillon et al., 2007), which is highly relevant for marital immigrants who may already face prejudice from local employers (Kim et al., 2020).

Second, living in high-accessibility areas can streamline the job search process for foreign spouses. Individuals closer to job-dense areas are likely to have more information about openings within their immediate neighborhood than those who live farther away and may be more aware of the advertisements posted in local shops or through word-of-mouth (Ihlanfeldt, 1997; Ihlanfeldt and Sjoquist, 1990). In contrast, marital immigrants, especially those who have just arrived in Taiwan, might struggle with inefficient job searches in unfamiliar territories and may find it challenging to identify openings due to a lack of knowledge about where to look or how local hiring processes work (Gobillon et al., 2011; Gobillon et al., 2007).

However, proximity alone may not be sufficient if local job opportunities do not align with the skills of foreign spouses or if nearby employers are reluctant to hire immigrants. In such cases, better ease of reaching opportunities through transit connectivity can expand the catchment area for employment and offer marital immigrants a broader range of potential options. Using the analogy of a lottery: the greater the number of applications a jobseeker can submit or interviews they can attend, the higher their chances of securing employment. In this context, transit connectivity enhances job search effectiveness by increasing the geographical area within which immigrants can look for work and increase the chances of a job-skills match. For foreign spouses with limited proficiency in Chinese, an intensified job search may provide a better chance of finding employers who are more accommodating of their language limitations or finding positions in more immigrant-friendly areas. This dynamic is not confined to those seeking to enter the workforce. Part-time workers may be more motivated to take on full-time positions when they discover opportunities that match their availability and are reachable within a reasonable time using public transit.

Additionally, transit-based job accessibility is relevant for foreign spouses, as immigrants from various contexts tend to depend more on public transit than native-born populations (Allen et al., 2021; Chatman and Klein, 2009; Heisz and Schellenberg, 2004). Studies have shown that transit accessibility to jobs positively affects transit mode share for work-to-home trips (Allen et al., 2021). A reliable public transit system can thus encourage foreign spouses to reach work locations conveniently without the financial and logistical burdens of owning and maintaining a private vehicle. Being distant and disconnected from opportunities can lower the effective wage after commuting costs, sometimes to a level below what is acceptable for the worker (Gobillon et al., 2011). The

ability to rely on public transit may thus help remove barriers by lowering the costs of job searches and enabling foreign spouses pursue employment opportunities they might otherwise forego due to high commuting expenses.

Another crucial factor is competition. In this paper, we use a competitive accessibility metric, which penalizes accessibility levels based on competition from other jobseekers. This recognizes that marital immigrants may not fully benefit from living in job-dense areas if they face substantial competition from nearby jobseekers (Parks, 2004), particularly natives who, on average, may have higher qualifications or are inherently preferred by local employers (Kim et al., 2020).

H2. Transit-based job accessibility positively affects personal income.

In terms of salary, we hypothesize that higher transit-based job accessibility contributes to an increased likelihood of moving to a higher income bracket. Greater transit-based job accessibility opens up a wider range of job opportunities, and jobseekers may select the option which offers the highest salary (Gobillon et al., 2011; Lin et al., 2016). This applies not only to unemployed individuals. Research has shown that women, in particular, are often underemployed due to commuting constraints, as limited transit can limit them to accepting lower-paying jobs close to home (Hanson and Pratt, 2003). Therefore, those who may have initially accepted low-paying jobs can shift to better-paying positions after having been presented with more choices. Such mechanisms may be more relevant to immigrant spouses in the suburbs and exurbs, where a wider catchment area can be the determining factor towards employment in the secondary or tertiary sectors, which typically offer higher wages.

H3. Transit-based job accessibility positively affects the likelihood of above-minimum-wage earnings.

We also hypothesize that higher transit-based job accessibility is associated with an increased likelihood of earning above the minimum wage. The minimum wage is a socially recognized threshold of basic income, and for foreign spouses in Taiwan with multiple disadvantages, surpassing the minimum wage would represent a meaningful step towards economic integration. This hypothesis is supported by *sufficiency* principles, particularly Frankfurt's doctrine of sufficiency, which argues that "if everyone had enough, it would be of no moral consequence whether some had more than others" (Frankfurt, 1988).

Several mechanisms may explain this hypothesis. First, married women in Taiwan, due to family-based restrictions, are likely to engage in part-time or informal work through family-based businesses or self-employment, which often operates under less formal contracts and lower wage guarantees (Lu, 1992). Improved transit-based job accessibility may allow foreign spouses to work for more hours or find employment that allows enough working hours to surpass the minimum wage. Second, it is also possible that immigrant spouses in job-dense areas can have more exposure to formal recruitment channels which are expected to advertise positions that are more likely to adhere to wage regulations and exceed the minimum wage threshold. Finally, transit-based job accessibility can also facilitate a shift from primary to secondary or tertiary sectors, which are more likely to offer salaries above the minimum wage.

H4. The impact of transit-based job accessibility varies across metropolitan and regional settings.

Finally, we hypothesize that transit-based job accessibility affects labor market outcomes differently across metropolitan and regional settings. We presume that this can be explained by the variations in land use patterns and transport systems.

One plausible reason is the difference in job density and diversity. In metropolitan areas, higher job density means that even within a relatively small geographic area, a slight increase in accessibility can open up a substantial number of job locations. This is not only applicable in terms of the volume of jobs but also in the diversity of opportunities

across multiple industries, which can be particularly important for addressing job-skills mismatches among immigrant spouses. Metropolis also tend to have more developed transport networks, where a train or bus ride can allow foreign spouses to conveniently reach less congested parts of the city where competition for jobs may be lower. In regional areas, however, job opportunities are often more dispersed and existing transit options (e.g., bus routes with limited transfer options and high headways) may only yield limited gains in both the volume and diversity of available positions. In some of the outlying areas, transit routes are limited to radial corridors linking peripheral neighborhoods to/from urban cores and science parks.

Industrial composition may also explain these differences. While metropolitan areas may feature a wider array of industries, marital immigrants in Taiwan may not have the qualifications needed for upper-tier positions. This means that salary outcomes may be fairly homogeneous and that increasing the number of reachable jobs may not be enough to push earnings significantly up to a higher income bracket (i.e., a “ceiling” effect on earnings may be present) despite the availability of diverse job opportunities. In contrast, regional settings typically have a higher concentration of primary sector jobs. In this case, improved transit-based job accessibility could serve as the gateway to higher-paying secondary or tertiary sector positions in science parks or manufacturing zones.

4. Materials

4.1. Study area

We conducted analyses at two distinct spatial scales: Taichung City, a predominantly urban metropolis and the second largest city in Taiwan in terms of population, and the Central Taiwan region, characterized by a combination of rural and urban areas. According to the Council for Economic Planning and Development, Central Taiwan encompasses the following counties and city: Miaoli County, Taichung City, Changhua County, Nantou County, and Yunlin County. Collectively, this region accounts for approximately 24 % of the national population and covers 29 % of the national land area.

It is important to note that our choice of these spatial scales, metropolitan and regional, is deliberate and is reflective of Taiwan's administrative reorganization and evolving city-regional planning strategies. For instance, in 2010, the original urban core of Taichung City was consolidated with surrounding districts of the former Taichung County. This reflects a broader shift from a traditional urban-rural dichotomy towards more integrated regional models, such as “hub-and-spoke” configurations and polycentric development (Huang, 2021). As such, labor markets and commuting patterns are expected to cross traditional urban-rural boundaries. Individuals in rural or suburban neighborhoods may regularly commute into urban centers or industrial hubs dispersed across the region. Our metropolitan-regional framing thus aims to more accurately capture these changing spatial dynamics and ensure that our job accessibility measures and analyses reflect spatial interdependencies rather than considering artificially segmented labor markets, as in the case of urban-rural dichotomy.

Taiwan's administrative divisions under a city or county are structured into Lis (village/neighborhood) and Qus (district/township). Li is the smallest administrative unit in Taiwan, while Qu is the next hierarchical level and is thus composed of different Lis. Taichung City is composed of 625 Lis (average working-age population per Li: 3290 individuals; average land area per Li: 3.5 sq. km) and 29 Qus (average working-age population per Qu: 70,900 individuals; average land area per Qu: 76.2 sq. km). Central Taiwan, by comparison, consists of 2138 Lis (average working-age population per Li: 1985 individuals; average land area per Li: 5 sq. km) and 106 Qus (average working-age population per Qu: 40,000 individuals; average land area per Qu: 100 sq. km).

4.2. Data and samples

We obtained all individual-level data from the 2018 *Survey of New Immigrants Living Needs*, which is conducted every five years since 2003.

To focus specifically on the influence of accessibility on employment status, we have deliberately excluded individuals who are not part of the labor force (i.e., economically inactive individuals). We define the labor force as individuals who are either employed or actively seeking employment. This approach deviates from conventional research practices, which often consider employed versus non-employed individuals (including those not actively seeking work due to factors such as old age, pursuit of studies, or domestic work). The most common reasons for foreign spouses not engaged in the labor force include childcare, caring for family members who need assistance in daily life, and pregnancy-related matters (National Immigration Agency, 2020). These individuals have explicitly indicated that they are not currently seeking employment, and thus, accessibility improvements may not have a meaningful or immediate impact.

We obtained a total of 3129 foreign spouses who belonged to the labor force in the Central Taiwan sample and 1303 observations in the Taichung City sample. In the analyses of income levels (i.e., testing hypotheses H2 and H3), we deduced the sample to full-time and part-time employed foreign spouses who provided their income information in the survey. This resulted in a total of 2949 valid observations for the Central Taiwan sample, and 1208 valid responses for the Taichung City sample.

4.3. Variables

To test H1 (employment status), we defined the dependent variable with a three-level ordered configuration: unemployed, part-time employed, and full-time employed. For H2 (personal income) and H3 (likelihood of earning above the minimum wage), we relied on self-reported income data from foreign spouses. It is worth noting that the actual income levels of foreign spouses remain unknown as the self-reported incomes are provided in interval ranges. The dependent variable for H2 was structured as a three-level ordered configuration stratified into the following salary ranges: less than NTD 20,000, NTD 20,000–30,000, and greater than NTD 30,000. For H3, we used a binary configuration that cuts at the nearest value to the minimum wage when the survey was conducted (NTD 20,000).

For the independent variables, we adopt a two-level hierarchical structure, with the area-level variable as the level-2 variable, and individual-level variables as the level-1 variables. This hierarchical structure is visualized in Fig. 1(a), while the description and summary of the expected effects of all variables are listed in Table 1.

4.3.1. Area-level independent variable

We considered transit-based job accessibility as the area-level variable in our econometric analyses. Accessibility was first measured at the Li level. One limitation of this study is that we cannot match the estimated accessibility at the Li level with the residence of the survey respondent. This is because the NIA only provides residential data at the Qu level, as mandated by the law. To address this limitation, we calculated the population-weighted mean accessibility levels for each Qu and incorporated these values in our analyses to align with the available residential data of each foreign spouse.

4.3.2. Individual-level independent variables

The individual-level variables serve as controls, and are divided into three main categories: human and social capital, family background, and immigration background.

In the context of human and social capital, females traditionally face lower employment prospects compared to males. This is primarily due to a higher share of domestic responsibilities, especially in the case of foreign spouses. Individuals with a college diploma or technician

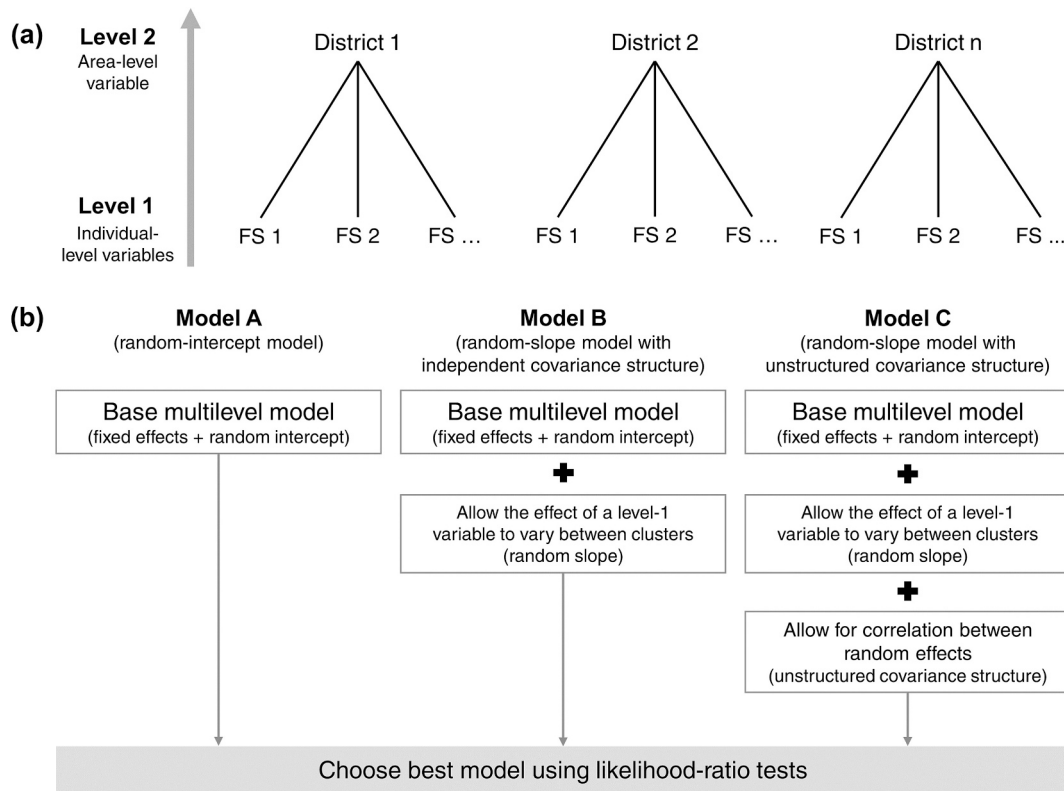


Fig. 1. Multilevel structure (a) and modelling strategy to determine the best model (b).
Note: FS = foreign spouse.

certificate possess advantages in the job market, which is expected to lead to better employment prospects. Additionally, proficiency in Chinese language skills is important in the day-to-day operations of Taiwanese companies, thereby expected to have a positive influence when engaging in the labor market.

The exact ages of the survey respondents are undisclosed, and are only reported in the form of age ranges. To account for this, we used dummy variables to represent the youngest and oldest age groups. This discretized configuration, however, poses challenges in predicting age effects on employment outcomes. The youngest individuals (aged 15–34) may experience favorable employment prospects due to their vitality and technological proficiency, but they may also face challenges tied to early family responsibilities. Likewise, the oldest group (aged 55 or above) may benefit from their extensive experience but could encounter health-related limitations affecting employment prospects.

In terms of family background, we anticipate that the number of young children will negatively affect employment probabilities. Foreign spouses with more grade school children (or below) typically need to allocate more time to childcare, limiting their availability for employment opportunities.

To mitigate potential endogeneity concerns, we deliberately omitted the independent variable related to family background when testing H2 and H3 (Table 1). This decision was made considering the potential bias from reverse association between these variables. For instance, a larger number of young children may prompt marital immigrants to pursue higher-paying employment to meet increased financial demands, while higher income could subsequently facilitate the support of additional children.

Finally, in terms of immigration background, we expect that employment assimilation will be observed among foreign spouses in Taiwan. We assume that the longer marital immigrants stay, the more they are able to adapt to the language and culture, acquire additional

labor market information, and consequently, improve their employment prospects. In terms of ethnicity, Southeast Asians traditionally have higher employment rates than those from Mainland China (Chuang et al., 2010). Similarly, previous editions of the survey revealed that Southeast Asians tend to be employed in the industrial sectors, and as such, we expect them to earn higher salaries.

4.4. Descriptive statistics

Table 2 shows the descriptive statistics for the dependent variables across both study areas. For H1 (employment status), which includes respondents in the labor force, a slightly higher proportion of sampled foreign spouses in Central Taiwan (92.75 %) were employed full-time compared to those in the Taichung City (90.64 %) sample. Unemployment rates were marginally lower in Central Taiwan (0.86 %) than in the Taichung City (1.46 %) sample.

For H2 (personal income) and H3 (above-minimum-wage earnings), which include employed respondents who reported income, the majority of foreign spouses in both samples reported earnings within the NT\$ 20,000–30,000 range. A slightly larger proportion of employed foreign spouses in Taichung City (76.49 %) surpassed the minimum wage threshold compared to those in the Central Taiwan (74.47 %) sample.

Table 3 presents the summary statistics for the independent variables. The mean transit-based job accessibility levels of foreign spouses in the Taichung City ($\bar{A}_i = 0.56$) sample were substantially higher than those in the Central Taiwan ($\bar{A}_i = 0.44$) sample. This discrepancy is expected given the concentration of job opportunities within urban areas and the availability of better public transport infrastructure in Taichung City. Conversely, at the regional level, remote underserved areas are prevalent. For instance, a considerable number of areas outside the county-administered cities in Nantou County and Yunlin County exhibit extremely low levels of transit-based job accessibility (Fig. 2(a)).

Table 1

Description and hypothesized effects of variables.

Independent variable	Description	Hypothesized effects			Supporting references
		H1 Employment status	H2 Personal income	H3 Above-minimum-wage earnings	
<i>Area-level variables</i>					
Transit-based job accessibility	Population-weighted mean accessibility levels	+	+	+	Kawabata (2003), Parks (2004), Yi (2006), Korsu and Wenglenski (2010), Matas et al. (2010), Sandoval et al. (2011), Lin et al. (2016), Blumenberg and Pierce (2017), Chen et al. (2017), Di Paolo et al. (2017), Haddad and Barufi (2017), Hu (2017), Blumenberg et al. (2019), Hernandez et al. (2020), Duarte et al. (2023a), Duarte et al. (2023b), Bastiaanssen et al. (2022), Bastiaanssen et al. (2025)
<i>Human and social capital</i>					
Age (15–34 years)	1 if the respondent is aged 15 to 34 years	+/–	+/–	+/–	Kawabata (2003), Parks (2004), Korsu and Wenglenski (2010), Matas et al. (2010), Sandoval et al. (2011), Blumenberg and Pierce (2014), Quintanar (2015), Blumenberg and Pierce (2017), Hu (2017), Blumenberg et al. (2019), Hernandez et al. (2020), Bastiaanssen et al. (2022), Duarte et al. (2023a), Duarte et al. (2023b), Bastiaanssen et al. (2025)
Age (55 or above)	1 if the respondent is aged 55 or above	+/–	+/–	+/–	Kawabata (2003), Parks (2004), Korsu and Wenglenski (2010), Matas et al. (2010), Sandoval et al. (2011), Blumenberg and Pierce (2014), Quintanar (2015), Blumenberg and Pierce (2017), Hu (2017), Blumenberg et al. (2019), Hernandez et al. (2020), Bastiaanssen et al. (2022), Duarte et al. (2023a), Duarte et al. (2023b), Bastiaanssen et al. (2025)
Female	1 if the respondent is female	–	–	–	Kawabata (2003), Yi (2006), Korsu and Wenglenski (2010), Lin et al. (2016), Hu (2017), Blumenberg and Pierce (2017), Duarte et al. (2023a), Bastiaanssen et al. (2025)
College graduate	1 if the respondent graduated from college	+	+	+	Bania et al. (2008), Korsu and Wenglenski (2010), Matas et al. (2010), Chuang et al. (2010), Blumenberg and Pierce (2014), Quintanar (2015), Lin et al. (2016), Di Paolo et al. (2017), Hernandez et al. (2020), Duarte et al. (2023a), Duarte et al. (2023b)
Good Chinese skills	1 if the respondent reported to have ‘good’ Chinese proficiency across all aspects - listening, speaking, reading, and writing	+	+	+	Parks (2004), Yi (2006), Chuang et al. (2010)
Technician certificate	1 if the respondent received a technician skills certification (either Class A, Class B, or Class C)	+	+	+	Chuang et al. (2010)
<i>Family background</i>					
Young children (age < 10)	Number of children younger than 10 years old	–			Parks (2004), Korsu and Wenglenski (2010), Matas et al. (2010), Sandoval et al. (2011), Chuang et al. (2010), Quintanar (2015), Hu (2017), Di Paolo et al. (2017), Duarte et al. (2023a), Bastiaanssen et al. (2022), Bastiaanssen et al. (2025)
<i>Immigration background</i>					
Length of stay in Taiwan	Number of years since migrating to Taiwan	+	+	+	Chuang et al. (2010)
Southeast Asian	1 if the respondent’s country of origin is a Southeast Asian country	+	+	+	Chuang et al. (2010)

Note: (+) denotes a positive effect, (-) denotes a negative effect, and (+/-) denotes that the effect is uncertain.

In terms of individual-level data, survey respondents were predominantly female, accounting for more than 90 % of foreign spouses for both sets of samples. Approximately one out of four respondents belonged to the youngest age group. Additionally, at least four out of ten respondents in both sets claimed to possess a good command of the Chinese language. On average, the length of stay in Taiwan for respondents was slightly over 12 years. There exists a slightly higher proportion of Southeast Asian spouses in Central Taiwan than in Taichung City.

5. Methods

5.1. Measuring transit-based job accessibility

We measured transit travel times using the population-weighted centroids of every Li as the origins and destinations of trips. These centroids were determined based on the 2015 Global Human Settlement Layers population grid. We performed travel time estimations using *r5r*, an open-source R package for transport routing (Pereira et al., 2021). Our routable transit network consisted of General Transit Feed Specification (GTFS) data for public transport modes with fixed routes and schedules, such as city buses and conventional railways, combined with

Table 2

Descriptive statistics of dependent variables.

Dependent variable		Central Taiwan		Taichung City	
		Count	%	Count	%
H1	Employment status	3129	100.00	1303	100.00
1	Full-time employed ^a	2902	92.75	1181	90.64
2	Part-time employed ^b	200	6.39	103	7.90
3	Unemployed ^c	27	0.86	19	1.46
H2	Personal income ^d	2949	100.00	1208	100.00
1	Greater than NTD 30,000	493	16.72	236	19.54
2	NTD 20,000–30,000	1703	57.75	688	56.95
3	Less than NTD 20,000	753	25.53	284	23.51
H3	Earning above the minimum wage	2949	100.00	1208	100.00
1	Greater than NTD 20,000	2196	74.47	924	76.49
2	Less than NTD 20,000	753	25.53	284	23.51

^a Full-time employed individuals are defined as those engaged in any form of work not conducted during spare hours, including those who are earning but temporarily unable to attend work during the survey period due to reasons such as injury or leave.

^b Part-time employed individuals are those engaged in temporary work during spare hours, typically after household responsibilities or during vacation.

^c Unemployed individuals are those not engaged in any form of work, but actively seeking employment and available for work, including those who have lost their jobs or voluntarily left work during the survey period.

^d 1.0 USD is approximately equal to 30 NTD.

road network and elevation data. To account for variations in public transport services, we computed travel time matrices for every minute during the peak morning period (from 7 am to 9 am) and used median travel times for our accessibility estimates.

We then adopt a competitive accessibility metric, where we consider both supply (job opportunities) and demand (working-age population) aspects, reflecting the reality that people compete for a limited number of jobs. In the employment context, competitive accessibility measures are better predictors of labor market outcomes than accessibility metrics which do not consider demand (Merlin and Hu, 2017). Specifically, we utilized the Balanced Floating Catchment Area (BFCA) method, which employs normalized weights to allocate demand and level of service proportionally (Paez et al., 2019), as shown in Eq. (1) below:

$$A_i = BFCA_i = \frac{\sum_{j=1}^J O_j [f(c_{ij}) w_{ij}^i]}{\sum_{i=1}^I P_i [f(c_{ji}) w_{ji}^i]} \quad (1)$$

where A_i is the accessibility level of demand node i using BFCA, O_j is the number of jobs in supply node j , P_i is the working-age population at

demand node i . The normalized weights w_{ij}^i assigns the working-age population proportionally to supply points, while w_{ji}^j assigns the level of service proportionally to the demand points. A more detailed discussion of the BFCA method can be found in Paez et al. (2019).

For this research, we defined $f(c_{ij})$ and $f(c_{ji})$ as binary travel time impedance functions, which assign a value of 1 if the travel time, c_{ij} , between zones i and j is within the threshold time t_{max} , and 0 otherwise. The binary configuration is applicable for employment scenarios, as we presume that job selection is primarily influenced by factors such as salary, job responsibilities, or company reputation, rather than solely travel time. The discretized configuration of the decay function suggests that in reality, people generally tolerate minor variations in travel time to and from work, as long as it remains within a reasonable threshold (Luo and Qi, 2009).

To address potential biases associated with the Modifiable Temporal Unit Problem (MTUP)—where the arbitrary choice of travel time cutoffs can influence results—we adopted the time interval modelling approach introduced by Tomasiello et al. (2023). Instead of setting a single cutoff t_{max} , we measure accessibility for every minute within a given time interval, and get the median access levels per Li. The formula for the interval-based BFCA is shown as follows:

$$A_i = IBFCA_{iT} = \text{median} (\{BFCA_{it} \forall t \in T\}) \quad (2)$$

where $IBFCA_{iT}$ is the median BFCA levels of demand node i within the travel time interval T , and T is the minute-by-minute distribution of threshold time within a given time interval from t_{min} to t_{max} . For this research, we set an interval of $T = [15, 45 \text{ min}]$. In the absence of up-to-date commuting data, we relied on previous research findings for reference. According to Ministry of Labor statistics from 2001, the average one-way commute time to work in Central Taiwan was 22 min, while the national average was 27 min. Additionally, Lien et al. (2020) reported that most high school students in Taichung City spend 10 to 50 min commuting to or from school, with very few exceeding an hour.

Note that we conducted separate estimations for each spatial scale. Therefore, accessibility levels for Taichung City in Fig. 2(a) and Fig. 2(b) may have minor disparities, especially in Lis near the borders with neighboring counties. The implication in the regional context (Fig. 2(a)) is that jobseekers in Taichung City may tap into job opportunities in neighboring counties within their catchment area, and vice versa. In the metropolitan context (Fig. 2(b)), job opportunities and competition among jobs are confined within the city boundaries.

The estimation of accessibility levels was supported by the *accessibility* package in R (Pereira and Herszenhut, 2023). Population counts were retrieved from the Socio-Economic Geographic Information

Table 3

Descriptive statistics of independent variables.

Independent variables	Central Taiwan				Taichung City			
	Entire labor force		Employed labor force		Entire labor force		Employed labor force	
	Mean/%		Mean/%		Mean/%		Mean/%	
Area-level variables								
Transit-based job accessibility	0.440	(SD = 0.216)	0.440	(SD = 0.214)	0.565	(SD = 0.210)	0.565	(SD = 0.209)
Human and social capital								
Age (15–34 years)	25.5 %	(n = 797)	26.0 %	(n = 766)	25.5 %	(n = 332)	26.2 %	(n = 316)
Age (55 or above)	3.0 %	(n = 93)	3.0 %	(n = 89)	3.6 %	(n = 47)	3.7 %	(n = 45)
Female	93.8 %	(n = 2935)	94.2 %	(n = 2778)	92.6 %	(n = 1207)	93.3 %	(n = 1127)
College graduate	7.9 %	(n = 246)	7.3 %	(n = 215)	10.0 %	(n = 130)	9.3 %	(n = 112)
Good Chinese skills	41.5 %	(n = 1300)	41.3 %	(n = 1217)	44.7 %	(n = 582)	43.9 %	(n = 530)
Technician certificate	6.0 %	(n = 188)	6.0 %	(n = 176)	5.8 %	(n = 76)	5.6 %	(n = 68)
Family background								
Young children (age < 10)	0.668	(SD = 0.805)			0.647	(SD = 0.790)		
Immigration background								
Length of stay in Taiwan	12.379	(SD = 5.784)	12.375	(SD = 5.749)	12.057	(SD = 5.949)	12.046	(SD = 5.894)
Southeast Asian	47.9 %	(n = 1498)	48.9 %	(n = 1442)	42.9 %	(n = 559)	44.6 %	(n = 539)
Number of observations	3129		2949		1303		1208	
Number of clusters (districts)	100		100		25		25	

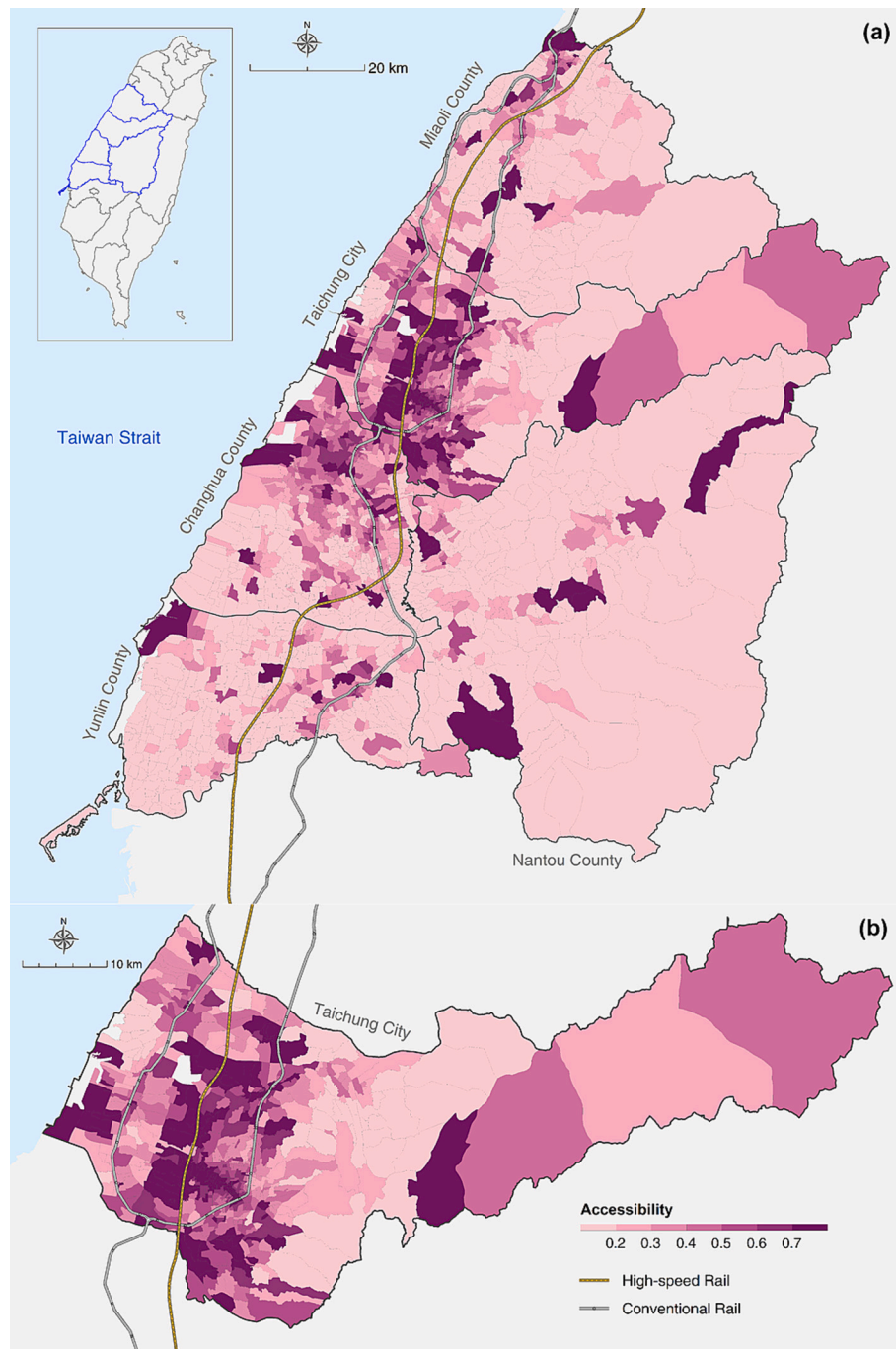


Fig. 2. Accessibility estimates per Li at the regional (a) and metropolitan (b) scales.

System (SEGIS) database of the Ministry of the Interior, while job distribution was obtained from the latest Industry and Service Census. Given our study's focus on employment, we only considered individuals who belonged to the working-age population (aged 15–64).

In essence, the accessibility variable derived in this study represents the total level of service of employment that is reachable within a reasonable travel time for each demand node. This metric accounts for competition and potential biases associated with the MTUP.

5.2. Multilevel mixed-effects logistic model

We developed multilevel mixed-effects logistic models to deal with the area-level and individual-level variables. This approach addresses violations of the independence assumption of generalized linear models,

thereby minimizing potential biases in standard error estimates (Dupont et al., 2013; Park et al., 2017).

In Eq. (3), we posit the existence of a continuous latent variable Y_{ij}^* underlying the observed discrete outcome Y_{ij} . For the analysis of dichotomous outcomes (testing H3), the latent variable configuration is demonstrated in Eq. (4). Eq. (5) shows the configuration for the analysis of ordinal outcomes (testing H1 and H2).

$$Y_{ij}^* = \gamma_{00} + \sum_{q=1}^Q \gamma_{0q} Z_{qj} + \sum_{p=1}^P \gamma_{p0} X_{pji} + \sum_{p=1}^P u_{pj} X_{pji} + u_{0j} + \varepsilon_{ij} \quad (3)$$

where,

Y_{ij}^* : continuous latent response for individual i in cluster j

γ_{00} : fixed intercept (or constant)

γ_{p0} : level-1 fixed slope (mean effect of a level-1 variable X_p in the overall sample)
 γ_{0q} : level-2 fixed slope (mean effect of a level-2 variable Z_q in the overall sample)
 X_{pij} : level-1 predictors
 Z_{qi} : level-2 predictors
 u_{pij} : random slope for level-1 variable X_p
 u_{0ij} : random intercept
 ε_{ij} : level-1 error term ($\varepsilon_{ij} \sim \text{logistic}(0, \pi^2/3)$)

$$Y_{ij} = \begin{cases} 0, Y_{ij}^* \leq 0 \\ 1, Y_{ij}^* > 0 \end{cases} \quad (4)$$

where,

$$Y_{ij} = \begin{cases} 0, Y_{ij}^* \leq \theta_1 \\ 1, \theta_1 < Y_{ij}^* \leq \theta_2 \\ 2, Y_{ij}^* > \theta_2 \end{cases} \quad (5)$$

where,

Y_{ij} : observed ordered response for individual i in cluster j
 θ_1, θ_2 : threshold parameters

We estimated the parameters using maximum likelihood estimation in Stata 18. We used the mean-variance adaptive Gauss-Hermite quadrature with 50 quadrature points for integration. Prior to estimations, we also centered all our predictors at the grand mean. Centering provides the intercepts and variances with meaningful implications, facilitates faster calculations, and reduces convergence problems. The following works provide detailed explanations of centering and other multilevel model assumptions: [Sommet and Morselli \(2017\)](#), [Skrondal and Rabe-Hesketh \(2010\)](#), and [Hox et al. \(2017\)](#).

5.3. Multilevel modelling strategy

In this section, we outline the multilevel modelling strategy that we used to guide the selection of the most suitable model. We decided not to estimate all random slope variance parameters, as this maximal approach carries the risk of overparameterization, convergence issues, and uninterpretable findings ([Bates et al., 2015](#); [Hox et al., 2017](#)). Instead, we adopt a more conservative approach and estimate the random slope variance parameter only when there are theoretical reasons to do so ([Hox et al., 2017](#); [Sommet and Morselli, 2017](#)).

Specifically, we investigated the potential variation in the effect of being Southeast Asian on labor market outcomes across districts. This decision was made to capture the potential effects of residing within immigrant enclaves, among others, as we lack data on the finer resolution of foreign spouses' residence. Previous literature has demonstrated that ethnic groups often co-locate to facilitate the sharing of social networks and resources, with different groups facing varying levels of access to employment-related resources and opportunities ([Blumenberg and Smart, 2014](#)). Therefore, we hypothesize that the effect of being Southeast Asian may vary between districts. In areas with significant Southeast Asian populations and businesses, being Southeast Asian may be positively associated with employment prospects due to aligned job opportunities, while negatively correlated elsewhere.

[Fig. 1\(b\)](#) illustrates the three-tiered multilevel modelling approach employed in this study. We progress from one tier to the next only when the inclusion of the added term significantly improves the model fit. Initially, we construct a base multilevel model (Model A) comprised of all fixed effects coefficients (γ parameters) and the random intercept term (u_{0ij}). Next, we introduce the random slope parameter (u_{1ij}) for the level-1 dummy variable representing being Southeast Asian (Model B). Finally, we allowed the random effects to be correlated (Model C). The best model is determined through likelihood-ratio tests (see [Table A-1](#)).

Note that in the first two tiers, we utilized an independent covariance structure for the random terms (u_{0ij}, u_{1ij}), where we estimated one unique variance parameter per random effect, while all covariances $\text{cov}(u_{0ij}, u_{1ij})$, are set to 0 (Eq. (6)).

$$\Sigma = \text{Var} \begin{bmatrix} u_{0ij} \\ u_{1ij} \end{bmatrix} = \begin{bmatrix} \sigma_{u0}^2 & 0 \\ 0 & \sigma_{u1}^2 \end{bmatrix} \quad (6)$$

where, σ_{u0}^2 denotes the estimated variance parameter for the random intercept, while σ_{u1}^2 denotes the estimated variance parameter for the random slope.

To account for the correlation between random effects at the district level, we explore an unstructured covariance (Eq. (7)), allowing us to distinctly estimate all variances and covariances.

$$\Sigma = \text{Var} \begin{bmatrix} u_{0ij} \\ u_{1ij} \end{bmatrix} = \begin{bmatrix} \sigma_{u0}^2 & \sigma_{u0,u1} \\ \sigma_{u0,u1} & \sigma_{u1}^2 \end{bmatrix} \quad (7)$$

where, $\sigma_{u0,u1}$ denotes the covariance between the random effects.

6. Results and discussion

6.1. Model estimation results

We estimated two distinct models for each labor market outcome—one for the Central Taiwan sample and another for the Taichung City sample. The empirical findings are summarized in [Table 4](#), which consolidates the results from [Table B-1 \(H1\)](#), [Table B-2 \(H2\)](#), and [Table B-3 \(H3\)](#). Prior to running the estimations, we assessed multicollinearity among the independent variables by calculating variance inflation factors (VIF). All variables had VIF values below 2, suggesting that multicollinearity was not a significant issue in any of the models (refer to Supplementary Material).

6.1.1. Influence of transit-based job accessibility on employment status (H1)

[Table B-1](#) presents the estimates from the multilevel ordered logit model used to examine the factors influencing employment status among economically active foreign spouses. Likelihood-ratio tests for both models indicate significant variability between districts, supporting the use of a multilevel model over a standard ordered logit model. The significant Wald χ^2 provides evidence to reject the null hypothesis that the slopes of our predictors are all zero, and that at least one of the predictors included in the model has a meaningful impact on the odds of the outcome occurring. For robustness, we re-estimated parameters using bootstrapping and employed robust (Hubert/White/sandwich) variance-covariance estimators. The consistency of results across these methods supports the robustness of our main interpretations (refer to Supplementary Materials).

Our empirical findings support H1 in the metropolitan context; that is, transit-based job accessibility ($\gamma = 2.022$; 95 % CI = [0.158, 3.887]; $z = 2.13$) positively affects employment status for foreign spouses. Specifically, a 0.1 increase in accessibility levels is associated with a 22.4 % increase in the odds of progressing to a higher employment status for a typical foreign spouse in Taichung City, holding all other factors constant. However, no significant relationship between transit-based job accessibility ($\gamma = 0.555$; 95 % CI = [-0.683, 1.795]; $z = 0.88$) and employment status was found at the regional scale.

Regarding human and social capital, we found that female spouses have 87.1 % lower odds of progressing to a higher employment status compared to their male counterparts in Taichung City, and 92.5 % lower odds in Central Taiwan, ceteris paribus. Having a college degree, technician certificate, or proficient Chinese language skills does not significantly affect employment status for foreign spouses in both sets of study areas.

In terms of family background, we found that each additional young child is associated with a 35.1 % decrease in the odds of advancing to a

Table 4

Summary of the effects of predictors on labor market outcomes of foreign spouses.

Variables	H1. Employment status		H2. Personal income		H3. Earning above the min. wage	
	Central Taiwan	Taichung City	Central Taiwan	Taichung City	Central Taiwan	Taichung City
District-level variable						
Transit-based job accessibility	o	+	+++	o	+++	+
Human and social capital						
Age (15–34 years)	o	o	+	++	o	o
Age (55 or above)	-	o	-	o	--	-
Female	---	--	---	---	---	--
College graduate	o	o	+++	+++	o	+
Good Chinese ability	o	o	+++	o	++	o
Technician certificate	o	o	++	++	+	o
Family background						
Young children (age < 10)	--	--				
Immigration background						
Length of stay in Taiwan	+++	+	+	+	+++	+++
Southeast Asian	++	+++	+	o	+	o

Note: +++ significant positive effect, $\alpha = 0.001$.++ significant positive effect, $\alpha = 0.01$.+ significant positive effect, $\alpha = 0.05$.

o no significant effect.

- significant negative effect, $\alpha = 0.05$.-- significant negative effect, $\alpha = 0.01$.--- significant negative effect, $\alpha = 0.001$.

higher employment status among foreign spouses in Taichung City, and a 24.9 % decrease in Central Taiwan, *ceteris paribus*. This suggests that parents' decision to engage in the labor market involves a careful consideration of time allocation between work and family-related responsibilities.

In terms of immigration background, we found that each additional year since migration to Taiwan is associated with a 5 % increase in the odds of progressing to a better employment status in Taichung City, and a 6 % increase in Central Taiwan, all else being equal. Moreover, the significant random slope for the Southeast Asian variable suggests significant variation in the effect of being Southeast Asian across districts, with the average fixed effect being positive and significant across both samples.

6.1.2. Influence of transit-based job accessibility on personal income (H2)

Table B-2 presents the estimates obtained from the multilevel ordered logit model used to analyze the factors influencing personal income levels among full-time and part-time employed foreign spouses. Similarly, the likelihood-ratio tests for both models suggest that the estimated multilevel models significantly outperform the single-level ordered logit.

Our empirical findings indicate that transit-based job accessibility ($\gamma = 0.985$; 95 % CI = [0.422, 1.548]; $z = 3.43$) significantly impacts personal income among foreign spouses in Central Taiwan, supporting H2 in the regional context. Specifically, we found that a 0.1 increase in job accessibility levels in Central Taiwan is associated with a 10 % increase in the odds of progressing into a higher income level, *ceteris paribus*. However, no significant relationship between transit-based job accessibility ($\gamma = 0.237$; 95 % CI = [-0.621, 1.095]; $z = 0.54$) and income is observed among foreign spouses in Taichung City.

In terms of the control variables, being female is consistently negatively correlated with income, while being part of the youngest age group (aged 15–34) is associated with better income prospects in both sets of samples. Similarly, possessing a college degree or a technician certificate correlates positively with higher income levels among foreign spouses. This suggests that while attaining a college degree (or technician certificate) may not guarantee a better employment status in Taiwan, it may facilitate progression from one income level to another, especially considering that such qualifications are often prerequisites for upper-level job positions.

In terms of immigration background, we found that the length of stay

in Taiwan is significantly associated with better income in both sets of samples, while the effect of being Southeast Asian is only significant and positive in the regional context.

6.1.3. Influence of transit-based job accessibility on the likelihood of above-minimum-wage earnings (H3)

Table B-3 shows the estimates of the multilevel binary logit model, which we used to examine the factors influencing the likelihood of earning above the minimum wage. The likelihood-ratio test was found to be significant only in the Central Taiwan sample. For the Taichung City model, using a single-level configuration would yield similar performance to a multilevel model.

Our empirical findings provide evidence that transit-based job accessibility has a significant and positive impact on the likelihood of above-minimum-wage earnings among foreign spouses in both metropolitan ($\gamma = 0.905$; 95 % CI = [0.043, 1.767]; $z = 2.06$) and regional ($\gamma = 1.377$; 95 % CI = [0.690, 2.064]; $z = 3.93$) contexts, supporting H3 in both sets of samples. Specifically, we found that a 0.1 increase in job accessibility levels is associated with a 9.5 % increase in the odds of above-minimum-wage earnings for a typical foreign spouse in Taichung City, and a 14.8 % increase in Central Taiwan, all else being equal.

In terms of the control variables, we found being female and being part of the oldest age group (aged 55 or above) to be negatively correlated with the likelihood of earning above the minimum wage. Good Chinese language ability and having a technician certificate are associated with better prospects of above-minimum-wage earnings in Central Taiwan. On the other hand, having a college degree emerged as a significant positive predictor within the Taichung City sample.

Regarding immigration background, we found similar relationships to hold true as in H2: longer stays in Taiwan correlate to better likelihood of above-minimum-wage earnings, while being Southeast Asian is a significant positive predictor solely within the Central Taiwan context.

6.2. Discussion

In the urban context, we found evidence that supports hypotheses H1 and H3. It is important to acknowledge that some foreign spouses are constrained by household duties, which may limit their employment prospects. By enhancing the ease of reaching jobs, the time previously spent on commuting can now be allocated to other valuable activities, enabling them to engage more fully in the labor market. Additionally,

improved job accessibility may provide them with access to jobs that accommodate their limitations, which would otherwise be out of reach with lower accessibility levels. If job accessibility is a determining factor in the transition towards better employment statuses, similar forces may enable foreign spouses to progressing towards earning above the minimum wage. This relationship is intuitive, as securing a better employment status (i.e., full-time job) provides a stable means of income and likely guarantees above-minimum-wage earnings.

However, our empirical evidence does not support H2; that is, job accessibility is not a significant predictor of personal income in Taichung City. One possible reason for this is the relative homogeneity in salary levels in the urban context. In the Taichung City sample, foreign spouses already live in areas with at least intermediate job access levels, with 98.3 % of surveyed spouses employed in the secondary or tertiary sectors. Expanding the catchment area within job-rich city centers may not necessarily lead to job opportunities with significantly higher salaries. This is especially true given that foreign spouses often face qualification constraints and social biases that may restrict their advancement to higher-level or managerial roles, thereby creating a potential income “ceiling.” While salary heterogeneity certainly exists among secondary and tertiary sectors, the difference in income levels within these sectors may not be as significant as the potential income increase that could be achieved when transitioning out of traditionally lower-paying primary sector jobs.

On the other hand, we found evidence supporting hypotheses H2 and H3 in the regional context. Note that there exist more sampled foreign spouses in Central Taiwan who live in areas with relatively lower job accessibility. Additionally, a higher proportion (11.4 %) of surveyed foreign spouses in Central Taiwan are employed in the primary sector. Thus, the statistical significance of H2 and H3 suggest that job accessibility may help foreign spouses in the suburbs and exurbs transition from primary employment to accessing secondary and tertiary industries in core urban areas or science parks, where salaries are traditionally higher and more likely to exceed the minimum wage. For robustness, we empirically verified these independent relationships: that is, job accessibility is significantly associated with higher likelihood of engaging in the secondary or tertiary sector, while employment in these sectors positively correlate with income and likelihood of above-minimum-wage earnings (refer to Supplementary Material). However, disentangling the direct and indirect effects of job accessibility and the potential mediating effect of secondary or tertiary sector employment on labor market outcomes lie beyond the scope of this paper.

Meanwhile, we found no evidence of job accessibility significantly influencing employment status among marital immigrants at the regional scale. This may be due to a higher reliance on private means of transport in transit-poor districts, suggesting that motorized vehicle ownership or accessibility may be stronger predictors of employment status. Moreover, there is also the possibility of a higher volume of

Table 5

Summary of findings in relation to existing literature.

Literature	Study area	Population	Mode	Effect of job accessibility on labor market outcomes		
				Employment probability/status ^a	Salary	Earning above the minimum wage
Kawabata (2003)	US	Low-skilled	Car, transit	(+) highly auto-dependent areas		
Parks (2004)	US	Immigrant women	Car	(+) native-born black, foreign-born Mexican and Vietnamese women		
Yi (2006)	US	General	Transit	(+) entire sample		
Bania et al. (2008)	US	Welfare	Car, transit	(o)	(o)	
Korsu and Wenglenski (2010)	France	General	Car, transit	(+) entire sample		
Matas et al. (2010)	Spain	Female	Transit	(+) entire sample		
Sandoval et al. (2011)	US	Welfare	Car, transit	(+) Los Angeles (transit); Alameda (walk-and-ride, drive alone)		
Blumenberg and Pierce (2014)	US	Low-income	Car, transit	(+) entire sample ^b		
Quintanar (2015)	Mexico	General	Car, transit	(+) women		
Lin et al. (2016)	Taiwan	Aborigines	Car, transit, motorcycle	(o)	(+) entire sample	
Blumenberg and Pierce (2017)	US	Welfare	Car	(+) entire sample		
Chen et al. (2017)	Ghana	General	Car, paratransit	(+) males and females		
Di Paolo et al. (2017)	Spain	General	Transit	(+) females		
Haddad and Barufi (2017)	Brazil	General	Car, transit		(+) entire sample	
Hu (2017)	US	General	Car	(+) medium-to-low-income groups		
Blumenberg et al. (2019)	US	Older adults	Transit	(+) entire sample		
Hernandez et al. (2020)	Uruguay	General	Transit	(+) entire sample		
Bastiaanssen et al. (2022)	UK	General	Transit	(+) low-income groups, transit-dependent urban areas		
Duarte et al. (2023a)	Brazil	General	Not specified		(+) entire sample ^c	
Duarte et al. (2023b)	Brazil	General	Not specified	(+) males and females		
Bastiaanssen et al. (2025)	Netherlands	General	Public transport-and-bicycle	(+) entire sample		
This study	Taiwan	Marital immigrants	Transit	(+) Taichung City (o) Central Taiwan	(o) Taichung City (+) Central Taiwan	(+) Taichung City (+) Central Taiwan

Note: (+) significant positive relationship; (o) no significant relationship; (–) significant negative relationship.

^a (+) includes negative relationship between poor job accessibility and employment prospects.

^b Found that moving to a neighborhood with better public transit accessibility is associated with employment retention, but not with job gains.

^c Found that better job accessibility is associated with lower likelihood of being a low-wage worker.

informal or family-based jobs in regional settings, where other factors, such as localized social networks or family connections, may be stronger predictors of securing a better employment status (e.g., moving from unemployment to part-time employment). Further investigation into these relationships is warranted for a more comprehensive understanding.

We then present a baseline summary of our study's findings in relation to existing evidence using Table 5. Note that our aim is not to offer direct comparisons, rather to provide a more comprehensive understanding of existing evidence, given the diverse contexts and methodologies employed in different studies.

We found similarities with research conducted in the European context, where a significant association was observed between better job accessibility and better employment prospects (Bastiaansen et al., 2022; Matas et al., 2010). Taichung City exhibits similar characteristics of dense and mixed land use, with presumed higher reliance on transit among disadvantaged groups. On the other hand, studies conducted in sparsely populated and less transit-dependent areas in the US found that car-based job accessibility appears to be a stronger predictor of employment probabilities (Blumenberg and Pierce, 2017). This may partially explain why transit-based accessibility may not be the determining factor in helping foreign spouses overcome barriers to employment in the regional context.

Previous research has paid less attention to the linkages between job accessibility and salary. In the Taiwanese context, Lin et al. (2016) found evidence of a positive relationship between car-based job accessibility and salary among urban aborigines. These findings partially align with our results, suggesting that car-based accessibility, rather than transit, may be a stronger predictor of income among urban minorities.

To the best of the authors' knowledge, this study is one of the first to explore the relationship between job accessibility and the likelihood of above-minimum-wage earnings. Our findings may have profound implications in the realm of empirical transport justice, suggesting that transit-based accessibility may play a transformative role in preventing foreign spouses from feeling socially isolated by providing them with opportunities to achieve a sufficient standard of living.

Furthermore, we contribute to a more extensive and up-to-date understanding of the labor market activity of foreign spouses in Taiwan, building on the findings of Chuang et al. (2010). Note that our study differs in terms of the employment definition used, as they were only able to classify employment as employed or non-employed due to survey limitations at that time (2003). Nonetheless, we found similar evidence that the number of young children consistently has a negative impact on employment prospects, while possessing a college degree is not a significant predictor. Additionally, both of our studies observed assimilation effects, where spouses who have resided longer in Taiwan experience better outcomes.

7. Conclusions and recommendations

7.1. Conclusions

Transport can exacerbate existing socio-economic disadvantages faced by marital immigrants or help them overcome barriers and engage better in the labor market. In this paper, we investigate the influence of transit-based job accessibility on the following labor market outcomes among foreign spouses in Taiwan: employment status (H1), personal income (H2), and the likelihood of above-minimum-wage earnings (H3). We then explore how these impacts differ across metropolitan and regional settings.

Using government survey data and multilevel econometric models developed, we demonstrate that better transit-based job accessibility is significantly associated with better employment status and higher likelihood of above-minimum-wage earnings in Taichung City. In Central Taiwan, we found that accessibility is a significant positive predictor of income levels and the likelihood of earning above the minimum wage.

These findings suggest that job accessibility may support the economic integration and income sufficiency of foreign spouses, albeit through distinct, context-specific pathways. In urban areas with already decent accessibility levels, improved job accessibility can expand the volume and diversity of available opportunities and potentially support transitions into better employment statuses, which are likely to provide enough working hours and formal protections consistent with minimum wage requirements. In the regional context, we provide evidence that better job accessibility is associated with upward economic mobility by enabling the pursuit of jobs in higher-paying industries, which would otherwise be out of reach with low accessibility levels. Thus, our empirical findings challenge the idea that job accessibility alone offers a silver bullet and that the impacts depend on the local socio-spatial contexts in which it operates.

Our work advances existing literature through the following empirical and methodological contributions. Firstly, our study is among the few to examine these dynamics in a non-Western setting. By focusing on the unique circumstances of foreign spouses in Taiwan, we illustrate how cultural norms, patterns of urban development, and migration contexts can contribute to labor market vulnerabilities that are not captured in literature dominated by Western case studies. In doing so, we contribute to a more globally inclusive understanding of transport accessibility and its potential implications for marginalized populations. Secondly, we provide insights into labor market outcomes that have received limited attention, such as salary. To the best of the authors' knowledge, this is among the first to explore the linkages between job accessibility and the likelihood of above-minimum-wage earnings. Thirdly, our research provides novel evidence across varying spatial scales, highlighting that the role of job accessibility in the largest metropolises cannot be directly applied to the regional context with its diverse contexts. Lastly, we depart from the majority of studies by using multilevel models, which aim to address the complexities of hierarchical data structures and mitigate potential statistical biases. Our proposed modelling strategy may also serve as a template for future research aiming to improve the robustness of multilevel estimates.

7.2. Policy implications

Our findings indicate that while individual factors remain important drivers of labor market outcomes, equitable land use and transport policies could help facilitate the economic integration of marital immigrants. Recognizing that job accessibility does not offer a one-size-fits-all solution to address all social issues among foreign spouses, we propose the following targeted policy implications.

In urban areas, interventions may be directed towards improving the existing public transport system by increasing the frequency of bus routes or extending operating hours, particularly on routes heavily utilized by unemployed and part-time employed foreign spouses. Land use policies may involve the provision of affordable housing in high accessibility neighborhoods or ensuring that marital immigrant families in core urban areas are protected from rent escalation ahead of the impending expansion of the metro. This reflects the core value of transit-oriented development in which affordable housing is one of crucial issues in metro system development in Taiwan.

In regional contexts, efforts may be directed towards establishing robust transport linkages between immigrant-rich areas with limited job accessibility and major commercial and industrial hubs. Alternative solutions such as demand-responsive transit services (DRTS), shuttle bus services, and vanpooling may support the city-to-suburb commutes, and vice versa, of low-income marital immigrants in areas where fixed transit routes may not be financially viable. In terms of land use, policymakers may consider strategies that facilitate the reallocation of job opportunities through land use zoning and the provision of industrial incentives. By strategically bringing manufacturing, construction, and related jobs closer to suburban areas, low-income foreign spouses may have better access to higher-paying jobs, while reducing the need for lengthy work-home commutes. While some of these proposals may not be entirely novel, our findings emphasize the ongoing need for sustained efforts to ensure that foreign spouses can reach employment opportunities with ease.

7.3. Limitations and directions for future work

We acknowledge the following limitations and propose directions of future studies accordingly.

Firstly, our empirical strategy and use of observational cross-sectional data means that we explicitly refrain from claiming causal interpretations of our results. Throughout this paper, we consistently interpret our estimated γ parameters as indicators of association rather than causal effects, and we have deliberately avoided language suggesting otherwise. Future research may consider using an instrumental variable approach (see Bastiaanssen et al. (2022), Duarte et al. (2023a), and Bastiaanssen et al. (2025)) or longitudinal data (see Blumenberg and Pierce (2014) and Blumenberg and Pierce (2017)) to strengthen causal inference.

Secondly, limited data resolution due to legal constraints have limited the scope of our analyses. Finer-resolution residential data at the Li level could have increased the number of upper-level clusters and improved estimation accuracy in multilevel models. Similarly, specific country of origin of foreign spouses was not available. The limitation lies in the potential loss of information resulting from clustering these individuals into a single broader group (i.e., 'Southeast Asia'). Recognizing the heterogeneity across nationalities is crucial, as their socio-demographic contexts and ability to share social networks and resources differ significantly (Blumenberg and Smart, 2014). This prevented a more detailed analysis, similar to the one performed by Parks (2004), which found associations between job accessibility and lower unemployment among specific nationalities.

Thirdly, we consistently assumed equal treatment of all jobs. However, in reality, jobs vary in terms of the competencies they require; that is, the skillset of a foreign spouse and the jobs that are directly accessible may not be compatible. It is in this vein that previous research explored accessibility to low-skilled jobs as an indicator of labor market outcomes among disadvantaged groups (Kawabata, 2003; Parks, 2004). Unfortunately, we were unable to incorporate this approach due to data limitations within the Industry and Service Census (Directorate-General of Budget, Accounting and Statistics, 2016).

Fourthly, this study focuses exclusively on foreign spouses, and while our findings suggest that job accessibility may positively influence their labor market outcomes, we cannot confirm whether these associations also apply to other groups. Due to data limitations, we were unable to include direct comparisons with native-born citizens or other

populations. We hypothesize that native spouses, who typically face fewer structural barriers (such as language challenges, cultural unfamiliarity, or discrimination), may experience smaller gains from job accessibility. That is, native spouses may not benefit as significantly as foreign spouses from expanded spatial coverage of job opportunities, which can help the latter group reach workplaces that are more accommodating or less discriminatory. This is consistent with existing research suggesting that job accessibility appears to have a stronger impact on more vulnerable segments, such as women (Di Paolo et al., 2017; Duarte et al., 2023b; Quintanar, 2015). Instead, the employment outcomes of native spouses may be more strongly influenced by human capital factors, such as education, credentials, or prior work experience. These propositions remain theoretical, and we encourage future research to empirically test how accessibility effects vary across populations with distinct socioeconomic constraints.

Fifthly, while this study accounted for socio-demographic variables relevant to the research context, we did not explore intersectional effects to maintain model interpretability. However, variables such as gender, age, ethnicity, and household structure likely interact in complex ways to shape labor market outcomes. Future studies could benefit from an intersectional framework to provide a deeper understanding of the disadvantages faced by certain groups.

Lastly, private cars and motorcycles play a significant role in daily mobility in Taiwan, especially in areas with limited public transport infrastructures. Therefore, in certain regions, car and motorcycle accessibility may outweigh transit-based job accessibility in predicting labor market outcomes. However, this aspect was beyond the scope of our research. Future studies could adopt the framework used by Lin et al. (2016) to estimate car and motorcycle-based accessibility in the Taiwanese context.

CRediT authorship contribution statement

Joshua Philip Suarez: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S.K. Jason Chang:** Writing – review & editing, Supervision, Resources, Conceptualization. **Jen-Jia Lin:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Likelihood-ratio tests to determine the best model configuration

Table A-1

Likelihood-ratio test results.

H1. Central Taiwan			H1. Taichung City		
Base model	Nested within: Model B	Model C	Base model	Nested within: Model B	Model C
Model A	LR $\chi^2(1) = 19.97$ Prob > $\chi^2 = 0.0000$		Model A	LR $\chi^2(1) = 19.43$ Prob > $\chi^2 = 0.0000$	
Model B		LR $\chi^2(1) = 5.03$ Prob > $\chi^2 = 0.0248$	Model B		LR $\chi^2(1) = 11.18$ Prob > $\chi^2 = 0.0008$
Better model	Model B	Model C	Better model	Model B	Model C
H2. Central Taiwan			H2. Taichung City		
Base model	Nested within: Model B	Model C	Base model	Nested within: Model B	Model C
Model A	LR $\chi^2(1) = 9.23$ Prob > $\chi^2 = 0.0024$		Model A	LR $\chi^2(1) = 4.88$ Prob > $\chi^2 = 0.0272$	
Model B		LR $\chi^2(1) = 0.57$ Prob > $\chi^2 = 0.4512$	Model B		LR $\chi^2(1) = 0.04$ Prob > $\chi^2 = 0.8447$
Better model	Model B	Model B	Better model	Model B	Model B
H3. Central Taiwan			H3. Taichung City		
Base model	Nested within: Model B	Model C	Base model	Nested within: Model B	Model C
Model A	LR $\chi^2(1) = 4.06$ Prob > $\chi^2 = 0.0438$		Model A	LR $\chi^2(1) = 0.08$ Prob > $\chi^2 = 0.7720$	LR $\chi^2(2) = 0.11$ Prob > $\chi^2 = 0.9478$
Model B		LR $\chi^2(1) = 2.45$ Prob > $\chi^2 = 0.1173$	Model B		
Better model	Model B	Model B	Better model	Model A	Model A

Note: the best configuration for each model is highlighted in **bold** characters; Model A – random-intercept model, Model B – random-slope model with independent covariance structure, Model C – random-slope model with unstructured covariance structure.

Appendix B. Multilevel model estimates

Table B-1

Multilevel mixed-effects ordered logit estimates for explaining employment status (H1).

Variable	Central Taiwan				Taichung City			
	Coef.	Std. Error	z	Sig.	Coef.	Std. Error	z	Sig.
<i>Dependent variable</i>	2: full-time employed; 1: part-time employed; 0: unemployed							
<i>Area-level variable</i>								
Transit-based job accessibility	0.556	(0.632)	0.88		2.022	(0.951)	2.13	*
<i>Human and social capital</i>								
Age (15–34 years)	0.030	(0.194)	0.15		0.125	(0.273)	0.46	
Age (55 or above)	−0.848	(0.407)	−2.09	*	−0.547	(0.535)	−1.02	
Female	−2.589	(0.746)	−3.47	***	−2.045	(0.770)	−2.65	**
College graduate	−0.020	(0.285)	−0.07		−0.005	(0.343)	−0.01	
Good Chinese ability	0.286	(0.196)	1.46		0.432	(0.250)	1.73	
Technician certificate	0.095	(0.340)	0.28		0.758	(0.616)	1.23	
<i>Family background</i>								
Young children (age < 10)	−0.287	(0.095)	−3.01	**	−0.432	(0.135)	−3.21	**
<i>Immigration background</i>								
Length of stay in Taiwan	0.058	(0.016)	3.61	***	0.049	(0.022)	2.24	*
Southeast Asian	0.876	(0.297)	2.94	**	1.557	(0.409)	3.81	***
<i>Threshold parameters</i>								
Cutoff threshold 1	−5.712	(0.266)		***	−5.289	(0.342)		***
Cutoff threshold 2	−3.281	(0.178)		***	−3.066	(0.254)		***
<i>Random parameters (for mixed-effects)</i>								
Var (Constant, district level)	1.120	(0.317)		***	0.782	(0.000)		***
Var (Southeast Asian)	0.821	(0.409)		*	1.110	(0.000)		***
Cov (Constant, Southeast Asian)	0.644	(0.309)		*	0.932	(0.000)		***
Number of observations	3129				1303			
Number of clusters: districts	100				25			
Log-likelihood	−787.25				−379.39			
Wald chi-squared	$\chi^2 = 52.31$			***	$\chi^2 = 50.59$			***
Likelihood-ratio test vs single-level model	$\chi^2 = 164.74$			***	$\chi^2 = 110.52$			***

Significance: * $\alpha = 0.05$, ** $\alpha = 0.01$, *** $\alpha = 0.001$.

Table B-2

Multilevel mixed-effects ordered logit estimates for explaining personal income levels (H2).

Variable	Central Taiwan				Taichung City			
	Coef.	Std. Error	z	Sig.	Coef.	Std. Error	z	Sig.
<i>Dependent variable</i>	2: > NTD 30,000; 1: NTD 20,000 - 30,000; 0: < NTD 20,000							
<i>Area-level variable</i>								
Transit-based job accessibility	0.985	(0.287)	3.43	***	0.237	(0.438)	0.54	
<i>Human and social capital</i>								
Age (15–34 years)	0.208	(0.099)	2.10	*	0.446	(0.152)	2.93	**
Age (55 or above)	−0.494	(0.228)	−2.16	*	−0.500	(0.321)	−1.56	
Female	−2.216	(0.183)	−12.13	***	−2.093	(0.270)	−7.74	***
College graduate	0.715	(0.158)	4.52	***	1.024	(0.222)	4.61	***
Good Chinese ability	0.421	(0.110)	3.83	***	0.303	(0.164)	1.85	
Technician certificate	0.492	(0.164)	3.00	**	0.700	(0.261)	2.68	**
<i>Immigration background</i>								
Length of stay in Taiwan	0.018	(0.008)	2.33	*	0.028	(0.012)	2.42	*
Southeast Asian	0.309	(0.121)	2.56	*	0.349	(0.188)	1.85	
<i>Threshold parameters</i>								
Cutoff threshold 1	−1.128	(0.071)		***	−1.270	(0.107)		***
Cutoff threshold 2	1.872	(0.079)		***	1.614	(0.116)		***
<i>Random parameters (for mixed-effects)</i>								
Var (Constant, district level)	0.219	(0.061)		***	0.120	(0.067)		
Var (Southeast Asian)	0.200	(0.099)		*	0.169	(0.119)		
Number of observations	2949				1208			
Number of clusters: districts	100				25			
Log-likelihood	−2680.39				−1115.31			
Wald chi-squared	$\chi^2 = 210.86$			***	$\chi^2 = 100.01$			***
Likelihood-ratio test vs single-level model	$\chi^2 = 65.64$			***	$\chi^2 = 18.03$			***

Significance: * $\alpha = 0.05$, ** $\alpha = 0.01$, *** $\alpha = 0.001$.**Table B-3**

Multilevel mixed-effects binary logit estimates for explaining the likelihood of above-minimum-wage earnings (H3).

Variable	Central Taiwan				Taichung City			
	Coef.	Std. Error	z	Sig.	Coef.	Std. Error	z	Sig.
<i>Dependent variable</i>	1: > NTD 20,000 (above min. wage); 0: < NTD 20,000 (below min. wage)							
<i>Area-level variable</i>								
Transit-based job accessibility	1.377	(0.350)	3.93	***	0.905	(0.440)	2.06	*
<i>Human and social capital</i>								
Age (15–34 years)	0.163	(0.119)	1.38		0.343	(0.182)	1.88	
Age (55 or above)	−0.663	(0.255)	−2.60	**	−0.790	(0.346)	−2.29	*
Female	−1.367	(0.284)	−4.82	***	−1.055	(0.363)	−2.90	**
College graduate	0.221	(0.196)	1.13		0.580	(0.281)	2.06	*
Good Chinese ability	0.343	(0.130)	2.63	**	0.288	(0.188)	1.54	
Technician certificate	0.467	(0.216)	2.16	*	0.520	(0.348)	1.49	
<i>Immigration background</i>								
Length of stay in Taiwan	0.032	(0.009)	3.45	***	0.050	(0.014)	3.61	***
Southeast Asian	0.282	(0.140)	2.02	*	0.343	(0.191)	1.79	
Constant	1.128	(0.081)	13.88	***	1.222	(0.093)	13.15	***
<i>Random parameters (for mixed-effects)</i>								
Var (Constant, district level)	0.322	(0.092)		***	0.063	(0.074)		
Var (Southeast Asian)	0.220	(0.145)						
Number of observations	2949				1208			
Number of clusters: districts	100				25			
Log-likelihood	−1594.86				−637.37			
Wald chi-squared	$\chi^2 = 61.23$			***	$\chi^2 = 35.65$			***
Likelihood-ratio test vs single-level model	$\chi^2 = 54.72$			***	$\chi^2 = 1.34$			

Appendix C. Supplementary dataSupplementary data to this article can be found online at <https://doi.org/10.1016/j.jtrangeo.2025.104358>.**Data availability**

Data will be made available on request.

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